

Questions with Answer Keys

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Q1 - Chord and its Properties

The locus of centre of the circle described on any focal chord of a parabola $y^2 = 4ax$, is

(1) $y^2 = 2a(x - a)$

(2) $y^2 = 2ax$

(3) $y^2 = 4a(x - a)$

(4) $y^2 = 2a(x + a)$

Q2 - Chord and its Properties

If a chord of $y^2 = 4ax$ makes an angle α , $\alpha \in (0, \pi/4]$ with the positive direction of X -axis, then the minimum length of this focal chord is

(1) $2\sqrt{2} a$ units

(2) $4\sqrt{2} a$ units

(3) $8 a$ units

(4) $16 a$ units

Q3 - Chord and its Properties

Let A and B be two points on a parabola $y^2 = x$ with vertex V such that VA is perpendicular to VB and θ is the angle between the chord VA and the axis of the parabola. The value of $\frac{|VA|}{|VB|}$ is

(1) $\tan \theta$

(2) $\tan^3 \theta$

(3) $\cot^2 \theta$

(4) $\cot^3 \theta$

Q4 - Chord and its Properties

The locus of feet of perpendiculars drawn from vertex of the parabola $y^2 = 4ax$ upon all such chords of the parabola which subtends a right angle at the vertex, is

(1) $x^2 + y^2 - 4ax = 0$

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(2) $x^2 + y^2 - 2ax = 0$

(3) $x^2 + y^2 + 2ax = 0$

(4) $x^2 + y^2 + 4ax = 0$

Q5 - Chord and its Properties

If the distance of two points P and Q from the focus of a parabola $y^2 = 4ax$ are 4 and 9 respectively, then the distance of the point of intersection of tangents at P and Q from the focus, is

(1) 8

(2) 5

(3) 6

(4) 12

Q6 - Tangent and its Properties

The equation of tangent to the parabola $y = (x - 3)^2$ parallel to the chord joining the points $(3, 0)$ and $(4, 1)$, is

(1) $2x - 2y + 6 = 0$

(2) $2x - 2y - 6 = 0$

(3) $4x - 4y + 11 = 0$

(4) $4x - 4y = 13$

Q7 - Tangent and its Properties

The angle between the tangents drawn from a point $(-a, 2a)$ to $y^2 = 4ax$ is

(1) $\pi/4$

(2) $\pi/2$

(3) $\pi/3$

(4) $\pi/6$

Q8 - Tangent and its Properties

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PN is an ordinate of the parabola $y^2 = 4ax$. A straight line is drawn parallel to the axis to bisect NP and meets the curve in Q . NQ meets the tangent at the vertex A in a point T such that $AT = k \cdot NP$, then the value of k is

- (1) $3/2$
- (2) $2/3$
- (3) 1
- (4) None of these

Q9 - Tangent and its Properties

AB, AC are tangents to a parabola $y^2 = 4ax$. p_1, p_2 and p_3 are the lengths of the perpendiculars from A, B and C respectively on any tangent to the curve, then p_2, p_1, p_3 are in

- (1) AP
- (2) GP
- (3) HP
- (4) None of these

Q10 - Tangent and its Properties

T is a point on the tangent to a parabola $y^2 = 4ax$ at its point P . TL and TN are the perpendiculars on the focal radius SP and the directrix of the parabola respectively, then

- (1) $SC = 2(TN)$
- (2) $3(SL) = 2(TN)$
- (3) $SL = TN$
- (4) $2(SL) = 3(TN)$

Q11 - Tangent and its Properties

AP and BP are tangents to the parabola, $y^2 = 4x$ at A and B . If the chord AB passes through a fixed point $(-1, -1)$, then the equation of locus of P is

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(1) $y = 2(x - 1)$

(2) $y^2 = 2(x - 1)$

(3) $y = 2(x + 1)$

(4) $y^2 = 2(x + 1)$

Q12 - Tangent and its Properties

Tangent to the curve $y = x^2 + 6$ at a point $P(1, 7)$ touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ at a point Q . Then, the coordinates of Q are

(1) $(-6, -11)$

(2) $(-9, -13)$

(3) $(-10, -15)$

(4) $(-6, -7)$

Q13 - Normal and its Properties

The distance between a tangent to the parabola $y^2 = 4ax$ ($a > 0$) and the parallel normal with gradient 1, is

(1) $4a$ units

(2) $2\sqrt{2}a$ units

(3) $2a$ units

(4) $\sqrt{2}a$ units

Q14 - Normal and its Properties

The normal at the point $P(ap^2, 2ap)$ meets the parabola $y^2 = 4ax$ again at $Q(aq^2, 2aq)$ such that the line joining the origin to P and Q are at the right angle. Then, p^2 is equal to

(1) 1

(2) 2

(3) $\sqrt{2}$

(4) 4

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Q15 - Normal and its Properties

If a line $x - y = 1$ cuts the parabola $y^2 = 4x$ at the points A and B . Normals at A and B intersect at C , other than above two meet the parabola in D , then D is equal to

(1) (1,1)

(2) (2,2)

(3) (4,-4)

(4) (6,6)

Q16 - Normal and its Properties

The locus of the point of intersection of normals at the ends of parallel chords of gradient m of the parabola $y^2 = 4ax$, is

(1) $2xm^2 - ym^3 = 4a(2 + m^2)$

(2) $2xm^2 + ym^3 = 4a(2 + m^2)$

(3) $2xm + ym^2 = 4a(2 + m)$

(4) $2xm^2 + ym^3 = 4a(2 - m^2)$

Q17 - Normal and its Properties

Normals are drawn from the point P with slopes m_1, m_2, m_3 to the parabola $y^2 = 4x$. If locus of P with $m_2 m_3 = \alpha$ is a part of the parabola itself. Then, the value of α is

(1) 1

(2) 2

(3) 3

(4) 4

Q18 - Normal and its Properties

A normal is drawn to parabola $y^2 = 4ax$ at any point other than the vertex. If it cuts the parabola again at a point whose distance from the vertex is not less than

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(1) $4a$

(2) $4\sqrt{6}a$

(3) $2\sqrt{6}a$

(4) $3\sqrt{6}a$

Q19 - Normal and its Properties

The length of intercept on the normal at the point $(at^2, 2at)$ of the parabola $y^2 = 4ax$ made by circle which is described on a focal distance of the given point as diameter is

(1) $a\sqrt{1+t}$

(2) $a\sqrt{1+2t}$

(3) $a\sqrt{1+t^2}$

(4) $a\sqrt{1+t^4}$

Q20 - Normal and its Properties

The tangent and normal at $P(t)$, for all real positive t , to the parabola $y^2 = 4ax$ meet the axis of the parabola in T and G respectively, then the angle at which the tangent at P to the parabola is inclined to the tangent at P to the circle passing through the points P, T and G , is

(1) $\cot^{-1} t$

(2) $\cot^{-1} t^2$

(3) $\tan^{-1} t$

(4) $\tan^{-1} t^2$

Q21 - Normal and its Properties

PQ is a normal chord of the parabola $y^2 = 4ax$ at P , A being the vertex of the parabola. Through P a line is drawn parallel to AQ meeting the X -axis in R . Then, the length of AR is equal to

(1) the length of latusrectum

(2) the focal distance of point P

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(3) the twice the focal distance of point P (4) the distance of the point P from the directrix

Q22 - Miscellaneous

The locus of the point of intersection of two tangents to the parabola $y^2 = 4ax$ such that they include an angle of 45° , is

(1) $y^2 - 4ax = (a + x)^2$

(2) $y^2 - 4ax = (a - x)^2$

(3) $y^2 + 4ax = (a + x)^2$

(4) $y^2 + 4ax = (a - x)^2$

Q23 - Miscellaneous

The equation of chord of contact of tangents drawn from a point $(3,4)$ to the parabola $y^2 = 2x$, is

(1) $x + 4y + 3 = 0$

(2) $x - 4y - 3 = 0$

(3) $x + y - 7 = 0$

(4) $x - 4y + 3 = 0$

Q24 - Miscellaneous

Tangents are drawn at the point, where the line $lx + my + n = 0$ is intersected by the parabola $y^2 = 4ax$, then the point of intersection of tangents is

(1) $\left(\frac{n}{l}, -\frac{2am}{l}\right)$

(2) $\left(-\frac{n}{l}, \frac{2am}{l}\right)$

(3) $\left(-\frac{n}{l}, -\frac{2am}{l}\right)$

(4) None of these

Q25 - Miscellaneous

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The locus of the point of intersection of tangents drawn at the extremities of a normal chord of the parabola

$$y^2 = 4ax, \text{ is}$$

$$(1) y^2(x + 2a) + 4a^3 = 0$$

$$(2) y^2(x - 2a) + 4a^3 = 0$$

$$(3) y^2(x + 2a) - 4a^3 = 0$$

$$(4) y^2(x - 2a) - 4a^3 = 0$$

Q26 - Miscellaneous

The pole of the line $lx + my + n = 0$ with respect to the parabola $y^2 = 4ax$ is

$$(1) \left(\frac{n}{l}, -\frac{2am}{l} \right)$$

$$(2) \left(\frac{n}{l}, -\frac{2am}{l} \right)$$

$$(3) \left(\frac{n}{l}, -\frac{2am}{l} \right)$$

(4) None of these

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